

PantryPilot Configuration Management Exercise

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Submission Overview

This submission applies the configuration-management example document to the PantryPilot semester project using generic project files. The package demonstrates repository setup, controlled configuration items, branching and pull-request workflow, automated CI checks, and release baseline management.

Project Context

PantryPilot is a smart meal and grocery planner. Its controlled artifacts include source code, runtime configuration, project documentation, tests, CI workflow files, changelog entries, and release notes. The generic files in this submission are intentionally lightweight, but they show how the real project would be managed under Git/GitHub configuration management.

CM Levels Demonstrated

Level	What the example required	PantryPilot artifact
Beginner	Create a repository, add generic files, commit, and push.	README, VERSION, config/app_config.json, src/pantrypilot_app.py, tests/test_smoke.py.
Intermediate	Use branches, pull requests, review, and conflict-control practices.	docs/branching_strategy.md, .github/pull_request_template.md, docs/git_commands_used.md.
Expert	Use CI/CD workflow, tags, releases, and automated checks.	.github/workflows/ci.yml, CHANGELOG.md, releases/release_notes_v1.0.0.md.

Configuration Items

Configuration item	File or folder	CM purpose
Source baseline	src/	Keeps application behavior under version control.
Tests	tests/	Provides repeatable verification for changes.
Runtime config	config/	Separates controlled settings from code.
CM documentation	docs/	Defines branching, change-control, and audit rules.
Automation	.github/workflows/ci.yml	Runs tests for pushes and pull requests.
Release records	VERSION, CHANGELOG.md, releases/	Tracks approved baselines and release history.

Change-Control Workflow

- Create a feature or fix branch from main.
- Modify only the configuration items needed for the change.
- Run tests locally before opening a pull request.
- Use the pull-request template to document change purpose, affected files, test evidence, and rollback notes.
- Merge only after review and passing CI checks.
- Update changelog and tag a release when a new baseline is approved.

Verification

The generic source/config path was smoke-checked locally by loading the PantryPilot configuration and verifying the baseline release description. The included GitHub Actions workflow installs requirements and runs pytest in a GitHub-hosted environment.

Submission Contents

- PantryPilot_CM_Generic_Files.zip: complete generic CM package for upload.
- PantryPilot_CM_Generic_Files/: unpacked source folder for inspection.
- PantryPilot_CM_Submission_Zixuan_Liang.pdf: summary document.